# Removing Sensitive Sentences from Italian Expert Appraisals: Distillation, Shortcuts and Counterfactual Repair

**3-cfu Project Work in Natural Language Processing**

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## Abstract

Real-estate enforcement proceedings require expert appraisals, which often mention the health, criminal history, minors and family circumstances of the people involved, and the GDPR requires stripping them before sharing these documents. In this work, we train a cheap and fast model that removes sentences with sensitive facts about a living person. Without an annotated corpus, we distil the labels of LLM annotators under a fixed two-gate policy, verified on a human-labelled sample, into an mDeBERTa classifier, adding counterfactual pairs and a pairwise margin loss. The selected variant removes 371 of 372 sensitive sentences in unseen appraisals while deleting about 5% of the harmless ones, far above entity-based tools and a zero-shot 12B LLM, and unlike the first model it keeps most negated sentences, which must not be removed.

## 1 Introduction

Expert appraisals (*perizie*) are detailed technical documents that serve as the backbone of value estimation for Italian real-estate properties that have been assigned to be auctioned. A professional expert selected by the court describes the property, its legal and cadastral status, its value and often its occupants.

Because the expert reports what they find on site, appraisals might often contain sentences that describe the legal or health status of the owner, or that describe their family status, maybe referring to minors. These sentences represent a limit in processing such documents with automated systems like hosted LLMs, because under the GDPR data minimisation policy (European Parliament and Council of the European Union, 2016, art. 5(1)(b)–(c)) such facts must be removed in any case where reuse goes beyond the proceeding itself. We call this task *sensitive sentence removal* (SSR). SSR is an asymmetric task, where a missed sentence (a *leak*) comes at a higher cost, like a fine, than a deleted harmless sentence, which only costs utility. Therefore we aim at a high recall while still measuring how much harmless text is lost. Given the regulation and the value of this data, we impose two further metrics to this task:

- *cost*, namely how resource-intensive is the model responsible for this process, since we cannot send these data to third-party providers we must own the model or process them locally;
- *speed*: for processing documents at scale we must keep the latency of the model as low as possible.

Standard available de-identification tools solve a different problem. Recognisers such as Presidio (Microsoft, 2026) and span taggers such as GLiNER (Zaratiana et al., 2024) find *entities* like names, dates, places, or a span matching a label such as "health condition". In appraisals the party is often already masked, and the risk lies in the *fact* attached to an identifiable person. Work on text sanitisation (Sánchez and Batet, 2016) and anonymisation (Lison et al., 2021; Pilán et al., 2022) frames disclosure risk in similar terms, but offers neither Italian data nor our category policy. Zero-shot LLMs have proven to be very powerful in related tasks (Gilardi et al., 2023), however their cost and speed don't make them a viable solution.

We treat SSR as a multi-label sentence classification task. The policy is written as a decision tree in the system prompt that a powerful LLM uses to distil labels, which are then used to fine-tune a small encoder that runs locally on a laptop. This LLM is hosted, but it only sees masked sentences, and only to build the labels. We evaluate on 5,950 documents split by near-duplicate group, using five test sets, which measure recall on held-out positives, harmless deletion

with an inverse-probability-weighted (IPW) estimate (Horvitz and Thompson, 1952), recall on a fully labelled 100-document audit, leaks on out-of-domain decisions of the Italian Supreme Court (Corte di Cassazione), and robustness on minimally edited counterfactual twins. We compare five student variants with three families of baselines. Our main findings are:

- The distilled classifier removes 371 of 372 held-out positives. Presidio reaches 86.3% only by deleting 52.87% of harmless sentences; GLiNER-PII, with thresholds calibrated like ours, reaches 51.9% at 11.49%, and a zero-shot 12B LLM 62.9%.
- High in-domain recall hides a vocabulary shortcut, since the first model deletes 54/54 negated twins and 24/24 twins about deceased persons.
- Counterfactual pairs alone do not teach negation (3/54 kept); pairs combined with a pairwise margin loss do (39/54), while keeping in-domain recall above 99%.
- The selected variant, `it3m_c`, keeps `A0`'s recall and reduces harmless deletion from 7.25% to 5.14% and leaking Supreme Court decisions from 13 to 2 of 307. It was chosen after viewing the tests, and every reference label comes from an LLM (Section 6).

## 2 Background

**Legal framework.** Health data are a special category under GDPR art. 9(1), and data on criminal convictions and offences are regulated by art. 10 (European Parliament and Council of the European Union, 2016). Although minors and family matters are not special categories, the Italian privacy code requires omitting the identity of minors and of parties to family-law proceedings when judicial decisions are disseminated (Repubblica Italiana, 2003, art. 52(5)); we extend this to appraisals by analogy (Appendix A). Our policy is shaped by three points:

- pseudonymised data remain personal data (GDPR recital 26), so a party masked as `[nome]` or "l'esecutato" stays identifiable through the property;
- deceased persons are not covered by the GDPR (recital 27);
- legal persons are not covered by the GDPR either (recital 14).

**Anonymisation and sanitisation.** Anonymisation masks direct and quasi-identifiers so that a person cannot be re-identified (Lison et al., 2021; Pilán et al., 2022), and court-ruling pipelines follow this route (Niklaus et al., 2023). Sanitisation instead removes content that remains harmful even when the identity is masked (Sánchez and Batet, 2016). Our task is sanitisation at sentence granularity, on top of a rule-based identifier mask (Section 3).

**Available solutions.** Existing tools fall into four families. (i) *Rule- and NER-based de-identification*, such as Presidio (Microsoft, 2026), combines pattern recognisers (fiscal codes, IBANs, dates) with a spaCy NER model and returns typed entity spans; it finds identifiers, not facts, and in our text most identifiers are already masked. (ii) *Zero-shot span taggers*, such as GLiNER (Zaratiana et al., 2024), encode the text together with natural-language labels in a single bidirectional encoder and score candidate spans against each label, so a category such as "health condition" can be queried without training; they still mark mentions, so a negated or hypothetical mention is tagged like an asserted one. (iii) *Prompted LLMs* can apply a written policy zero-shot, at a high cost per sentence and, as we show in Section 6, with inconsistent final decisions. (iv) *Supervised anonymisation and sanitisation models* (Pilán et al., 2022; Niklaus et al., 2023) learn the target from annotated data, which does not exist for Italian appraisals and our category policy; we build it with an LLM teacher instead. For our model we chose mDeBERTa-v3 (He et al., 2023), the multilingual version of DeBERTaV3, due to its small weight (278M parameters) and because it is multilingual, so it covers Italian, while DeBERTaV3 encoders perform well on classification benchmarks; we did not compare it with other encoders.

**LLM labels and distillation.** Gilardi et al. (2023) have proved that LLMs can match or exceed crowd workers on text-annotation tasks, which makes them usable as teachers in cases like ours, when expert annotation is unavailable. We distil their labels into a smaller student (in the spirit of Hinton et al., 2015) and report the agreement between two teacher passes with Cohen's $\kappa$

(Cohen, 1960).

**Shortcuts and counterfactuals.** Models often learn surface cues that correlate with the label instead of the intended rule (Geirhos et al., 2020; McCoy et al., 2019). Minimally edited inputs whose label flips expose and reduce this reliance, whether used as contrast sets for evaluation (Gardner et al., 2020) or as counterfactually augmented training data (Kaushik et al., 2020). Our twins and margin loss build on this idea.

## 3 Data

**Corpus and splits.** Our corpus is composed of documents attached to judicial sale listings on the Portale delle Vendite Pubbliche (Links section), almost all for auctions scheduled in 2026: 3,917 expert appraisals, plus sale notices, court orders and valuations, which for simplicity we call appraisals in the rest of the report. The documents are mostly dated between 2023 and 2026. We reduce the original 7,464 source files to 5,950 by deduplicating normalised text. We group reissued versions and template copies using MinHash LSH (Broder, 1997) (5-word shingles, Jaccard $\geq 0.85$) into 5,845 groups, and whole groups go to train, validation or test by a seeded hash (60/20/20), giving 3,521, 1,230 and 1,199 documents. Splitting happens before any labelling, so no copy of a test document is ever seen in training. Documents have a median of about 180 sentences.

**Pre-processing.** A deterministic regex mask runs before every step, namely labelling, training, evaluation and baselines. It replaces fiscal codes, birth dates, up to three capitalised tokens after a title (Sig., Dott., Avv., ...) and contact data with placeholders such as `[nome]` (65,042 occurrences in 5,198 documents). Names without a title remain visible, so the mask reduces identifiability without removing it (full rules in the code, see the Links section). Text is split into sentences on `.;!?` or blank lines, with exceptions for single letters, 37 legal and technical abbreviations and lowercase continuations. Sentences longer than 700 characters are cut into overlapping windows, and pieces shorter than 25 characters are dropped; in the rest of the report a window counts as a sentence. The corpus contains about 920,000 sentences, but only a small part of them is labelled, since we train on about 12,000 labelled sentences (some of them synthetic, Section 5) and test on 100 whole documents (37,320 sentences) plus about 2,400 sentences picked on purpose (Table 1).

**Labelling policy.** We distinguish four categories: *salute* (health), *penale* (criminal matters), *minori* (minors) and *famiglia* (family and private-life circumstances). A sentence is positive for a category only if it satisfies both following criteria: (1) it concerns an identifiable living person, and (2) it positively asserts the sensitive fact. Negated records ("no pending charges"), hypothetical or generic statements, and facts about deceased persons are therefore kept. Parties masked as `[nome]`, *omissis* or "l'esecutato" count as identifiable (as pseudonymised data). The teacher prompt encodes these rules as a decision tree, and all our prompts are in the code (Links section).

**Labels.** We start by annotating our corpus using two local LLM annotators, Qwen3.6-27B (Yang et al., 2025) and Gemma 4 12B (4-bit, temperature 0), keeping only the cases where both are certain and agree. Text resulting from this process shows some classes are quite rare, we found there are 34 examples for health and 102 for criminal matters among 5,104 real rows. We therefore decided to add 2,591 synthetic rows written by local LLMs from 29 scenario plans, for 7,695 rows in total (33.7% synthetic), which we call A0-train. A further audit of 300 windows showed that these annotators over-labelled boilerplate. We mitigate this effect by labelling more data with a stronger model (GPT-6.1 Sol), from now on the *teacher*, and applying a two-gate policy as a frozen decision tree (subject, living, assertion, topic) in two blind passes, excluding labels that are uncertain. On 299 deliberately hard probe sentences, the two passes agree on 293 (98.0%, $\kappa = 0.949$).

**How we use the data.** We use the data in three roles. A0-train trains the first student, and the later rounds add new rows to it (the Teacher rows, the Edge suite, the CF pairs and Cass-train, described in Section 5). The validation split is used only to choose the thresholds. Finally, five test sets, kept apart from any training, measure the models (Table 1).

**Evaluation sets** (Table 1). Sensitive sentences are quite rare, since in audit100 they are only 158 out of 37,320, about one in 240. Therefore, in order to measure recall we have to search for them, and the set we find in this way contains no ordinary

| Set | Docs | Sentences | Measures |
|---|---|---|---|
| Held-out | 116 | 372 pos.[a] | recall, doc leak |
| IPW sample | 200 | 2,000[b] | harmless deletion |
| audit100 | 100 | 37,320 | recall (158 pos.) |
| Cassazione | 847 | 1,180 | decision leak[c] |
| Twins | 24[d] | 54 pairs | negation |

Table 1: Evaluation sets, all labelled by the teacher except audit100 (hosted LLMs with a blind quality check); details in Appendix F. [a]famiglia 369, minori 7, salute 2, penale 2; 257 of them come from two rounds of 150 candidates picked among the sentences that `A0` and `it1` scored highest, so we also report the 115 found by keywords only, which we call the keyword subset. [b]Drawn by score band from 56,094 sentences (Appendix E); 1,937 certain harmless. [c]Supreme Court decisions from 2021 to 2026; 307 of them with positives (per head, salute 67, penale 235, minori 68 and famiglia 173). [d]Source groups; also 24 deceased, 15 company-subject and 196 category-swap twins.

| System | Sees | Fires for category $c$ when |
|---|---|---|
| Presidio | sentence | any entity is found (category-agnostic) |
| GLiNER-PII | sentence | a span for label $c$ scores $\geq \tau_c$ |
| Gemma 12B | $\pm 2$ sentences | its own decision (raw), or our policy applied to its answers (composed) |
| Students | $\pm 200$ chars | head $c$ scores $\geq \tau_c$ |

Table 2: What each system sees and how it decides. The masked text, the sentence segmentation, the deletion rule and the test sets are shared.

sentences; to count false alarms we need instead a separate sample of ordinary sentences, taken from documents that no model and no choice has ever touched. Table 1 lists the five test sets, how we built them and what they measure. Since positives in appraisals are almost all *famiglia*, rare-class recall is tested mainly on Cassazione.

## 4 Approach

**Task and formulation.** We evaluate every system on the same kind of decision, for each sentence $s$ and category $c$, it decides if the whole $s$ must be removed for $c$, and $s$ is replaced whole by `[omissis]` if any category fires. The systems differ only in what they see of the document, according to their specifications (Table 2).

**Baselines.** We start by testing what is already available off the shelf, giving each system the input it was designed for. For *Presidio* 2.2 we use spaCy `it_core_news_lg` with the Italian recognisers, and we delete a sentence if any entity scores at least 0.5 ("any entity"). *GLiNER-PII* (`gliner_multi_pii-v1`, the multilingual PII checkpoint of GLiNER) instead receives one natural-language label per category, for example "health condition / disease / disability of a person". A category $c$ fires if its best span reaches a per-category threshold $\tau_c$. We use it zero-shot, without fine-tuning, choosing English labels over Italian ones on validation, since the model is multilingual and covers Italian, but its labels work better in English (305 vs 221 of the 541 validation positives within the budget). Both Presidio and GLiNER-PII tag entities or spans in whatever text they receive, so they see the target sentence alone, since including neighbouring sentences might skew the judgement. As a small LLM we chose Gemma 4 12B instruct (4-bit, run locally). We prompt it with the teacher's own decision-tree prompt, giving it up to two sentences of context on each side, at temperature 0 with JSON output. We read its answer in two ways. *Raw* takes its own final decision, while *composed* ignores it and applies our two gates in code to its intermediate answers (topic, subject, living, assertion).

**Student model.** As a student we chose mDeBERTa-v3-base (278M parameters). We fine-tuned it on the LLM labels, with a classification head on the `[CLS]` state and one sigmoid output per category $c$, plus a fifth "other categories" output inherited from early runs, which is never used for deletion. We encoded each sentence $s$ with its local context as `Prima:` `<200 chars>` `[TARGET]` $s$ `[/TARGET]` `Dopo:` `<200 chars>`, truncating to 384 tokens with the target kept whole (Figure 1). The score $p_c$ is the maximum sigmoid output of head $c$ over the sentence's windows, and $s$ is removed if $p_c \geq \tau_c$ for any $c$. The loss is binary cross-entropy with per-head positive weights (the negative/positive ratio, capped at 10). All our following iterations share this architecture. The two gates are not built into the architecture, instead the model must learn them from the labels.

**Counterfactual pairs.** In order to teach the gates rather than category words, we build a set of 741 real-derived counterfactual pairs from training groups only, the CF pairs. These pairs are built by editing one attribute of a real sentence while keeping the rest verbatim (e.g. "A carico

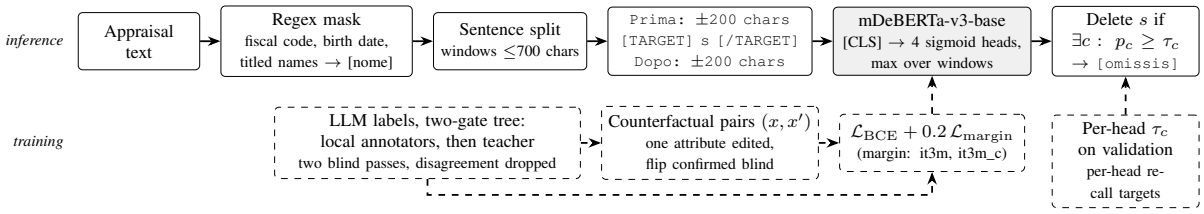



Figure 1: Pipeline. The top row shows inference on one masked appraisal, while the bottom row (dashed) shows how labels, counterfactual pairs, the loss and the per-head thresholds are produced.

dell'esecutato risultano carichi pendenti" becomes "A carico dell'esecutato non risultano carichi pendenti"). Thus we end up with negation (207), hypothetical (145), deceased subject (27), company subject (13) and category swap (349) pairs. A fresh blind pass keeps only the pairs whose label flips as intended (agreement 94.9%, $\kappa = 0.898$). An original sentence and its twin always fall in the same micro-batch, and a check confirms zero overlap with every test set. The test twins (Table 1) are built in the same way, from positives of documents that no model has seen (Appendix F).

**Margin loss.** Binary cross-entropy scores the two members of a pair separately. Since an original and its twin share almost all their words, the model can keep BCE low by giving them similar scores and relying on category words, and in fact with BCE alone `it3` keeps only 3 of 54 negated twins. We therefore add a hinge that asks the original to score at least a margin $m$ above its twin, for the keep-type pairs $P$ (negated, hypothetical, deceased and company twins):

$$\ell(x, x') = \frac{1}{|H_x|} \sum_{c \in H_x} \left[ m - z_c(x) + z_c(x') \right]_+$$
$$\mathcal{L}_{\text{margin}} = \frac{1}{|P|} \sum_{(x,x') \in P} \ell(x, x')$$

where $H_x$ are the heads positive for the original $x$, $z_c$ is the maximum logit of head $c$ over windows, $m = 1$, and the total loss is $\mathcal{L}_{\text{BCE}} + 0.2\,\mathcal{L}_{\text{margin}}$. We work on logits because near 1 the sigmoid flattens the gap, and category-swap pairs are excluded because both members must be removed.

**Implementation.** All training and inference ran locally on one laptop (Apple M4 Max, 36 GB, PyTorch on MPS), using the encoder, Presidio, GLiNER and the LLM off the shelf through `transformers`, `presidio-analyzer` and `gliner`. Our contributions are the task formulation, the labelling policy, the counterfactual pair construction, pair-aware batching, the margin loss, the threshold rules and the evaluation design, while most of the code was written by AI coding agents under our direction (see the AI Report).

## 5 Experiments

**`A0`.** We start by training `A0` on the 7,695 rows labelled by the two local annotators. We use AdamW (learning rate $2 \cdot 10^{-5}$, weight decay 0.01, 10% linear warm-up), an effective batch of 32 and seed 42 for 5 epochs, and we keep the checkpoint with the lowest validation loss (epoch 4). These are standard fine-tuning values, fixed in advance and not tuned.

**More teacher labels (`it1`).** `A0` deletes 7.25% of harmless sentences, therefore we first try to make it more precise by adding harder real negatives. `it1` continues the training of `A0` for 500 updates (learning rate $10^{-5}$), where half of each batch replays A0-train, so that the model does not forget what it learned, while the other half comes from the Teacher rows, 1,132 real rows labelled by the teacher, and from the Edge suite, 1,157 synthetic sentences built from templates around the hard cases of the policy, such as negations and hypotheticals. Since the new rows follow the final policy, `it1` also shows what better labels alone can change. Differently from the other students, `it1` uses one global threshold on the maximum head score, chosen on audit100.

**Counterfactual training (`it3`, `it3m`, `it3m_c`).** Looking at the errors of `A0` (Section 6) we found the model exhibits signs of a shortcut, reacting to category words instead of applying the two gates. In order to fix it, `it3` continues `A0` for 800 updates (learning rate $10^{-5}$). Half of each batch is again replay, while the other half mixes the Edge suite (40%), the CF pairs (35%), the Teacher rows (15%) and Cass-train, 500 Cassazione sentences rich in negations, taken from decisions that are not in the test set (10%). These proportions were set by hand, giving most of the new half to the

| Model | Held-out recall (/372) | IPW harmless deletion (%) | audit100 recall (/158) | Held-out doc leak (/116) | Cassazione dec. leak (/307) | Negated twins kept (/54) |
|---|---|---|---|---|---|---|
| Presidio, any entity | 321 (86.3%) | 52.87 | 122 (77.2%) | 33 (28.4%) | 77 (25.1%) | – |
| GLiNER-PII | 193 (51.9%) | 11.49 | 68 (43.0%) | 83 (71.6%) | 141 (45.9%) | – |
| Gemma 4 12B, composed | 234 (62.9%) | 0.00[§] | – | 65 (56.0%) | 158 (51.5%) | – |
| A0 | **371 (99.7%)** | 7.25 | **157 (99.4%)** | **1 (0.9%)** | 13 (4.2%) | 0 (0.0%) |
| it1[†] | 359 (96.5%) | 2.29 | 156 (98.7%) | 4 (3.4%) | 69 (22.5%) | 0 (0.0%) |
| it3 | **371 (99.7%)** | **1.72** | 155 (98.1%) | **1 (0.9%)** | 22 (7.2%) | 3 (5.6%) |
| it3m | 369 (99.2%) | 2.37 | 156 (98.7%) | 3 (2.6%) | 24 (7.8%) | **39 (72.2%)** |
| it3m_c | **371 (99.7%)** | 5.14 | **157 (99.4%)** | **1 (0.9%)** | **2 (0.7%)** | 32 (59.3%) |

Table 3: Main results on the test sets. 95% intervals for A0 and it3m_c: held-out recall 98.5–100% (both), IPW deletion 6.56–8.05% and 4.56–5.89%, Cassazione leaks 2.5–7.1% and 0.2–2.3%. [†]Single global threshold (0.060) chosen on audit100, so its audit100 values are optimistic. [§]On a weighted 600-sentence subsample, where A0 gives 7.27%. Baselines miss most originals, so their twins are omitted.

hard cases (Edge suite and CF pairs), and were not tuned. it3m is trained in the same way, with the margin loss on top (weight 0.2).

**Threshold calibration.** For the systems that output a score per category, namely GLiNER-PII and our students, we choose one threshold $\tau_c$ per category on validation. Each category has a recall target (salute 0.95, penale 0.85, minori 0.98, famiglia 0.98), and for A0 these targets give $\tau = (0.038, 0.407, 0.020, 0.072)$, which is A0's ratified operating point. it3 and it3m keep A0's thresholds. However, the margin compresses the penale scores, therefore it3m_c keeps the weights of it3m and only lowers $\tau_{\text{penale}}$ to 0.0098 (Appendix B). For GLiNER-PII we use the same targets, but we also require each category to fire on at most 5% of the 899 all-negative validation rows, so among the thresholds that respect both we take the one with the highest precision, and if none reaches the target we take the one with the highest recall within this budget.

**Metrics.** *Recall* is the share of positive sentences removed by any head, reported with a 95% confidence interval. A document *leaks* if at least one of its positives is kept. To know whether a model deletes too much, we need harmless sentences that no model and no choice has ever touched, therefore we define *harmless deletion* as the share of them that a model removes, measured on the 2,000 sentences of the IPW sample. These sentences are not drawn uniformly, so each one counts for as many sentences as it represents (details in Appendix E). Finally, to check whether two models really differ, we compare them on the same items and look only at the cases where one of them is right and the other is wrong, and if these cases are too unbalanced to be due to chance we consider the difference real.

**Decision rule.** A variant replaces A0 only if it has held-out recall $\geq 99\%$, audit100 recall $\geq 156/158$, no more held-out leaks than A0 and a lower IPW deletion than A0. We fixed this rule before it3m_c, but after seeing the results of it3 and it3m.

**Results.** Table 3 reports all models on the five test sets. Looking at the baselines first, Presidio is the one reaching the highest recall, however this only happens because it deletes 52.87% of the harmless sentences, so basically half of the text. GLiNER-PII, which we calibrated on validation in the same way as our students, reaches 51.9% at 11.49% harmless deletion, and none of its categories manages to reach the recall target while staying within the firing budget, while lowering a single threshold for all of them buys recall only with a massive deletion (95.7% recall at 71.63% deletion for $t = 0.05$, Appendix C). Gemma, if we take its own decision, removes almost nothing (3.5%), on the other hand composing its answers with our two gates raises the recall to 62.9% with no harmless deletion in the subsample, although more than half of the held-out documents (65/116) still leak.

Our students are all far above the baselines. A0 removes 371/372 held-out positives and 157/158 audit positives, paying for it with 7.25% harmless deletion. it1 trades recall for precision (96.5% recall, 2.29% deletion) and therefore fails the recall gate, and like A0 it keeps none of the 54 negated twins, which means that adding labels from the teacher was not enough to remove the shortcut. it3 is the one with the lowest deletion (1.72%),

but it misses three audit positives, leaks 22 Cassazione decisions and keeps only 3 negated twins, while adding the margin loss (`it3m`) brings the kept twins up to 39/54, at the price of the held-out leaks going up to 3/116. In the end `it3m_c` is the only variant satisfying all four criteria, in fact it equals `A0` on held-out recall, audit recall and held-out leaks (each model leaks one document, a different one, $p = 1.0$), and at the same time it deletes less harmless text (5.14% vs 7.25%), leaks fewer Cassazione decisions (2 vs 13, $p = 0.0034$) and keeps 32/54 negated twins instead of none. If we look only at the keyword subset (115 positives), which is not biased by the model scores, `A0` and `it3m_c` both reach 114/115, while Presidio gets 86.1% and GLiNER-PII 50.4%.

## 6 Discussion

**Baseline errors.** All the baselines look for entities, while in our task what matters is the fact attached to a person, whose name is often already masked. Most of Presidio's hits are places, organisations and dates, but in appraisals addresses and dates are exactly the content we want to keep. GLiNER's labels instead behave lexically, in fact its minori head finds almost every minor (144/147 on validation), since a minor is usually a mention ("figlio minorenne"), while salute finds only 12 of 42, since an illness is a fact that can be expressed in many ways. Gemma fails in a different way, namely it usually reads the facts correctly but then it does not carry them through to the final decision, in fact on Cassazione almost all of its harmless removals (173 of 176) concern subjects that it had itself labelled as hypothetical. Composing the decision in code fixes part of this, but not enough.

**Vocabulary shortcut.** We find that `A0`'s high recall does not mean that it learned the two gates. We found this by testing it on a keyword-mined frame of appraisals not used for training, where it deletes up to 40.2% of the *harmless* sentences containing salute, minori or penale keywords, against 7.5% of those without, and on the twins, where it deletes all 54 negated and all 24 deceased-subject twins. Part of this comes from its labels, which were made before the negation and deceased rules existed, but even `it1`, which also sees labels that follow these rules, still keeps no negated twin. We suspect that much of the shortcut comes from the training data, since one third of A0-train is synthetic, so the keywords in it are easy to learn. On the other hand, synthetic data is what makes the rare heads work at all, since students trained only on real rows had to delete more than half of the harmless audit sentences to reach 156/158 recall.

**Effect of the counterfactuals.** Adding the CF pairs to the training data was not enough on its own, since an original and its twin share almost all their words, so with BCE alone the model kept scoring them almost the same (`it3` kept only 3 of 54 negated twins). The margin term is what taught the model to look at the negation, because it penalises the difference between the two sentences rather than each score alone, while keeping in-domain recall above 99% (39 of 54 negated twins kept). Deceased subjects instead did not change at all, probably because we had too few deceased pairs for the model to learn from. Finally, lowering the penale threshold is a trade-off, since `it3m_c` recovers the Cassazione positives that `it3m` lost, but at the price of a few negated twins, which the penale head now removes too.

**Error analysis.** Table 4 lists the main error types. Our final model fails mostly on abstract statements, namely sentences that talk about a crime or an illness in general, like statements of law or hypotheticals, without referring to a real person. This is probably due to the training data, since in appraisals a crime or an illness is almost always attached to someone, so the model never really had to check who the sentence is about. On Cassazione, where decisions discuss crimes and illnesses in the abstract, `it3m_c` deletes 92.0% of the harmless sentences with penale keywords (79.1% for `A0`), and even in appraisals harmless sentences with keywords are deleted far more often than the others (up to 38.7%, against 4.8%). The margin therefore helped with negation, but it could not teach the model to check whether a keyword refers to a person, and for this adding abstract statements of law and hypotheticals to the training data would be the natural next step.

**Limitations.** (1) Every test reference is an LLM label, therefore one human annotator (the author) labelled 186 stratified sentences to check them. The human agrees with the teacher and with `A0`'s local annotators at a similar level, but on 49 sentences from `A0`'s old validation set much more with the teacher, since the local annotators overflag rare categories (Table 5). (2) The in-domain

| Error type | Invented example | Who |
|---|---|---|
| Keyword, no person | *Il locale è adibito ad ambulatorio medico.* | A0, it3m_c |
| Negated | *Non risultano carichi pendenti a carico del debitore.* | A0 (54/54) |
| Deceased | *Il bene pervenne per successione dal padre, deceduto nel 2001.* | all (24/24) |
| Law statement | *La legge prevede la sospensione della pena per grave malattia.* | it3m_c |
| Masked party | *L'esecutato è affetto da una grave invalidità.* | baselines |

Table 4: Error types with invented, anonymised examples. The first four are harmless sentences that get deleted (the law statement on Cassazione); the last is a positive that entity-based baselines keep.

| Comparison | $\kappa$ | Exact match |
|---|---|---|
| *119 sentences with both labels* | | |
| Human vs teacher | 0.80 | 88.2% |
| Human vs A0's local annotators | 0.82 | 89.9% |
| Teacher vs A0's local annotators | 0.92 | 95.0% |
| *49 sentences from A0's validation set* | | |
| Human vs teacher | 0.69 | 79.6% |
| Human vs A0's local annotators | 0.41 | 30.6% |

Table 5: Human agreement with the LLM labels on 119 sentences with both a teacher label and an agreed label from A0's local annotators (top), and on 49 sentences from A0's old validation set, all marked sensitive by the local annotators (bottom). $\kappa$ is pooled over sentences and categories; exact match means the same categories.

evaluation is basically a famiglia result (369 of 372 held-out positives), and the claims on rare classes rest on Cassazione. (3) We chose it3m_c among five variants after knowing their test results, since the shortcut could only be measured on the twins and on Cassazione, which have no validation counterpart. (4) Each model was trained once, with a single seed. (5) Every student starts from A0 and replays A0-train, so A0's older labels are part of every model, and a student trained from scratch on the final labels is left for future work.

## 7 Conclusion

In this work we tried to remove sensitive sentences from Italian expert appraisals with a small model that runs on a laptop, distilling into an mDeBERTa encoder a two-gate policy applied by LLM annotators. Our final model removes 371 of 372 held-out positives while deleting about 5% of the harmless sentences, which is far beyond what entity-based tools and a zero-shot 12B LLM can do, and with 279M parameters it processes about 45 sentences per second, while a 12B LLM is 150 to 529 times slower (Appendix D). The most interesting result for us was however a negative one, since a near-perfect in-domain recall was hiding a model that ignored negation and the living-person gate, and we only noticed it thanks to the counterfactual twins and the Supreme Court decisions. Counterfactual pairs alone did not fix this, while a margin loss on them taught the model to keep most negated twins. With one lowered threshold, our final model then leaks far fewer Cassazione decisions than A0 at the same in-domain recall, although out of domain it still deletes too many abstract statements.

The main limitations are the small human check and the post-hoc model selection (Section 6), so as next steps we would like to add a second annotator, more deceased and hypothetical pairs and several seeds.

## 8 AI Report

AI was used in two different roles.

**AI as part of the method.** LLMs produced all the reference labels, as described in Sections 3 and 4. Two local open models (Qwen3.6-27B, Gemma 4 12B) labelled A0's training data, and local LLMs wrote its synthetic rows. We used GPT-6.1 Sol as the teacher for all later labels, the counterfactual edits and the Cassazione labels. Hosted LLMs labelled and quality-checked audit100. Gemma 4 12B is also one of our baselines. We treat these outputs as data produced by models, not as human judgements.

**AI assistance.** AI coding agents (Claude Code and Codex) wrote most of the experiment code and ran most of the experiments under my direction. I defined the task, the labelling policy and its rulings, the evaluation design and every go/no-go decision. Claude drafted the teacher prompt from my written rulings, and the root-cause review of the failed retrains was carried out with both agents, before I ratified the resulting plan. During my manual validation the assistant explained the guideline on request and walked through 2 example items, which I excluded from the metrics. All numbers come from one canonical results file, regenerated by a script from aggregate outputs. AI was also

used to polish the language of the report.

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## 9 Links to external resources

- Code (including all prompts and the mask rules) and aggregate result files: https://github.com/klein-t/sensitive-sentence-removal-it. The corpus contains personal data and is not released.
- mDeBERTa-v3-base: https://huggingface.co/microsoft/mdeberta-v3-base

| Rule | Basis | Status |
|---|---|---|
| salute | art. 9(1) | law |
| penale | art. 10; C. 2-octies | law |
| minori | C. 52(5); rec. 38 | analogy |
| famiglia | C. 52(5) | analogy |
| Masked party: identifiable | rec. 26 | law |
| Hypothetical: keep | no subject | law |
| Deceased: keep | rec. 27 | law* |
| Negated: keep | art. 10 (arguably) | choice |
| Company seizure: remove | rec. 14 | choice |

Table 6: Which policy rules follow from the law and which are our design choices (art. and rec. are GDPR articles and recitals, C. are Codice privacy articles). minori and famiglia are not art. 9 categories; art. 52 covers judicial decisions and is applied by analogy. *The exception for an implicated living heir is our choice.

- GLiNER-PII: https://huggingface.co/urchade/gliner_multi_pii-v1
- Gemma 4 12B (run locally as a 4-bit MLX conversion): https://huggingface.co/google/gemma-4-12B-it
- Presidio: https://github.com/microsoft/presidio
- Portale delle Vendite Pubbliche (public judicial sale listings): https://pvp.giustizia.it
- Corte di Cassazione public decisions: https://www.italgiure.giustizia.it/sncass/

# 10 Appendix

## A. Policy rules and their legal basis

Table 6 shows which rules of our policy follow from the law and which ones are our own design choices, with all citations referring to Regulation (EU) 2016/679 (European Parliament and Council of the European Union, 2016) and to D.Lgs. 196/2003 as amended (Repubblica Italiana, 2003).

## B. Thresholds and validation

A0's thresholds for salute, penale, minori and famiglia are 0.0384, 0.40721, 0.02048 and 0.07239, and it3m_c keeps them except for penale, which we lowered to 0.0098325, since the margin compresses the penale scores and at A0's threshold the penale head of it3m almost never fires. Table 7 shows how the two models behave on validation.

| Head | A0 | it3m_c |
|---|---|---|
| salute | 6, 40/42 | 4, 35/42 |
| penale | 16, 46/53 | 43, 49/53 |
| minori | 3, 146/147 | 6, 145/147 |
| famiglia | 39, 293/299 | 28, 287/299 |

Table 7: Negatives fired (of 899) and recall on validation.

| System | Size | Device | Sent./s | ms/sent. |
|---|---|---|---|---|
| it3m_c | 279M | GPU | 45.5 | 22 |
| GLiNER-PII | 289M | GPU | 42.7 | 23 |
| Presidio | rules+CNN | CPU | 295.8 | 3 |
| Gemma 12B | 12B, 4-bit | GPU | 0.09–0.3 | 3.3k–11.6k |

Table 8: Size and speed of each system on one laptop (Apple M4 Max, 36 GB).

## C. Per-class results and GLiNER-PII

Both A0 and it3m_c remove every rare held-out positive, and on Cassazione it3m_c misses only two positives. GLiNER-PII instead cannot reach any recall target within the 5% budget, and a single global threshold does not help either, since it buys recall only with a massive deletion (95.7% recall at 71.63% deletion with $t = 0.05$).

## D. Speed and size

We measured the speed of each system end to end on 20 test documents, always on the same laptop (Table 8). it3m_c is about as fast as GLiNER-PII, Presidio is faster, and the 12B LLM is 150 to 529 times slower.

## E. Harmless-deletion estimate

We split the 56,094 sentences of the IPW documents into seven bands by the highest score of A0 and it1, and drew about 286 sentences per band, so that the rare uncertain sentences are well represented. Each sampled sentence then counts for as many sentences as it represents in its band, so harmless deletion becomes

$$\hat{D} = \frac{\sum_{i \in H} w_i \, \mathbf{1}[i \text{ deleted}]}{\sum_{i \in H} w_i}, \qquad w_i = \frac{N_{b(i)}}{n_{b(i)}}, \tag{1}$$

where $H$ is the set of sampled harmless sentences, $b(i)$ the band of sentence $i$, and $N_b$ and $n_b$ the sizes of band $b$ in all the documents and in the sample. This is the standard Horvitz-Thompson estimator (Horvitz and Thompson, 1952), and since the weights depend only on how we sampled, it holds for any model.

## F. Test sets in detail

*Held-out.* We looked for candidates in two rounds, in 338 documents drawn from the 868 that no model and no choice has ever used. In each round we took random sentences containing keywords, plus the 150 keyword sentences that `A0` and `it1` scored highest, and we labelled all of them with two teacher passes. Since the score-mined part, which gives 257 of the 372 positives, favours these two models, we also report the 115 positives found by keywords only, namely the keyword subset.

*IPW sample.* See Appendix E.

*Audit100.* We reserved 100 test documents and had them labelled sentence by sentence by hosted LLMs, then a larger LLM checked blind the sentences that contain a keyword or were flagged ($\kappa = 0.862$). Since this check only read these sentences, a positive missed by both labellers cannot be recovered.

*Cassazione.* We took public civil and criminal decisions of the Italian Supreme Court from 2021 to 2026 (see the Links section), found their sentences by keywords only, at most three per decision, and labelled them with two teacher passes, which agree on 98.4% of them ($\kappa = 0.964$). None of these decisions is in Cass-train.

*Twins.* Starting from real positives of unseen documents, the teacher edits one attribute (negation, deceased subject, company subject or category), leaving the context and the OCR noise as they are, and then a fresh blind pass keeps only the pairs whose label flips as intended.